

Amar Ali-bey

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🌐 <https://amaralibey.github.io>

ABOUT

- **Research and Publications** — Published in top AI venues: CVPR, BMVC, WACV, Neurocomputing.
- **Open-Source Contributions** — Created and maintained OpenVPRLab and GSV-Cities, a framework and dataset for large-scale place recognition (+1800 downloads, 500+ GitHub stars).
- **Deep Learning Expertise** — Image Geo-Localization, Face Recognition, Semantic Segmentation, Object Detection, LLM/LVLM.
- **Project Management** — Autonomous in full project lifecycle: data collection to model deployment.
- **Communication** — Effectively presents complex technical concepts at conferences and meetings (FR/EN).
- **Technical Skills** — PyTorch, Lightning AI, Tensorboard, Numpy, OpenCV, FAISS, Python, C/C++.

EDUCATION

Ph.D. in Computer Vision and Machine Learning

2017 — 2024

Laval University, Québec, Canada

Thesis title: *Deep Representation Learning for Visual Place Recognition.*

- Advisors: Prof. Brahim Chaib-draa and Prof. Philippe Giguère.

Master's in Computer Science

2015 — 2017

Laval University, Québec, Canada

Advanced courses: Mobile Robotics, Machine Learning, Optimization.

Explored multiple research topics: Image Retrieval, Object Detection and Semantic Segmentation.

Transitioned directly into Ph.D.

B.S. in Software Engineering (with honors)

2009 — 2014

Ecole nationale Supérieure d'Informatique (ESI), Algiers, Algeria

Attended Algeria's most competitive engineering school and *Graduated as Valedictorian.*

FIRST AUTHOR PUBLICATIONS

BoQ: A Place is Worth a Bag of Learnable Queries

(CVPR 2024)

Amar Ali-bey, Brahim Chaib-draa and Philippe Giguère

- Currently state-of-the-art on all existing benchmarks, with big margins.
- Paper, code, and trained models: <https://github.com/amaralibey/Bag-of-Queries>

MixVPR: Feature Mixing for Visual Place Recognition

(WACV 2023)

Amar Ali-bey, Brahim Chaib-draa and Philippe Giguère

- Paper, code, and trained models: <https://github.com/amaralibey/MixVPR>

Global Proxy-based Hard Mining for Visual Place Recognition

(BMVC 2022)

Amar Ali-bey, Brahim Chaib-draa and Philippe Giguère

GSV-Cities: Toward Appropriate Supervised Visual Place Recognition

(Neurocomputing 2021)

Amar Ali-bey, Brahim Chaib-draa and Philippe Giguère

- Paper and Dataset: <https://github.com/amaralibey/gsv-cities>

WORK EXPERIENCE

Deep Learning Consultant

2018 — 2024

Short/Medium term interventions with high impact on performance.

- Football players Re-Identification in video streams
- Traffic cone detection
- Semantic segmentation for urban scenes
- Helmet detection for safety in work sites
- Enhanced the deep learning pipeline for object tracking and counting

Graduate Teaching Assistant

2017 — 2023

Laval University

Prepared new materials for courses, assisted students with technical questions, and taught parts of the courses:

- **Advanced Techniques In Artificial Intelligence (IFT-4102/7025)** [100+ students]
- **Deep Learning (GLO-4030/7030)** [70+ students]
- **Practical Machine Learning (GLO-7050)** [50+ students]
- **Introduction to Mobile Robotics (GLO-4001/7021)** [80+ students]

The Autonomous Vehicle of Laval University (VAUL)

2018 — 2019

- Developed a cone detector using OpenCV to enhance the annotation of cones in circuit videos.
- Trained YOLO models for real-time cone detection in video streams.

Student Internship

2013 — 2014

University of Technology of Compiègne (UTC), Compiègne, France

Worked on the Vehicle Routing Problem with Time Windows (VRPTW), which consists of selecting the optimal routes for a fleet of vehicles to service customers within specific time frames. Under the supervision of Pr. Aziz Moukrim.

ACADEMIC ACHIEVEMENTS AND ACTIVITIES

1st Rank Awards

2012, 2013, 2014

Received annual prizes from DJEZZY, a school partner, for maintaining top of class.

Attended International Computer Vision Summer School (ICVSS)

2017

Volunteered at IEEE International Conference on Robotics and Automation (ICRA)

2019

Presented at Montreal AI Symposium

2022

Reviewer for many AI venues (ICRA, IROS, WACV, RA-L)

2020-2024

PROJECTS

OpenVPRLab

<https://github.com/amaralibey/OpenVPRLab>

A comprehensive open-source framework for SOTA visual place recognition (VPR).

- Highly modular design allowing focused development on specific VPR components.
- Easy dataset management with a single-command download for training and validation datasets.
- Model Zoo of existing VPR techniques.
- Integrated Tensorboard support for tracking experiments.
- Performance visualization tools for analyzing model retrievals.

GSV-Cities

<https://www.kaggle.com/datasets/amaralibey/gsv-cities>

A large-scale dataset for training deep neural networks for Visual Place Recognition (VPR).

- More than 500k images depicting over 60k different places spread across multiple global cities.
- Brings down training time from multiple days to sub 3 hours (by eliminating pair/triplet mining).
- Enhanced the performance of existing techniques by up to 1000%.

SKILLS

Programming languages: Python, C, C++, PHP, SQL.

Frameworks/Libraries: PyTorch, Numpy, OpenCV, LightningAI, FAISS, TensorBoard.

ML/AI: Instance Retrieval, Semantic Segmentation, Object Detection, Image Classification, LLM/LVM.

Communication Languages: Englis, French.